Asmi Umesh Pulgam

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Professional Summary

Software Engineer pursuing an M.S. in Computer Science with experience building scalable backend systems, distributed databases, cloud-native applications, and AI-powered software. Passionate about developing high-performance software that supports data-intensive and engineering-focused applications. Currently looking for Spring 2027 co-op opportunities and full time roles starting from May 2027.

Education

Arizona State University — Tempe, AZ, US
Master of Science in Computer Science - GPA: 4.00 / 4.00 — *Aug 2025 – May 2027*

Savitribai Phule Pune University — Pune, MH, India
Bachelor of Engineering in Computer Engineering, Honors in Data Science - GPA: 3.50 / 4.00 — *Aug 2021 – May 2025*

Technical Skills

Programming Languages: Python, Java, Rust, C++
Frontend: React.js, Next.js, HTML5, CSS3, Bootstrap
AI/ML: TensorFlow, PyTorch, LLM, LangChain, RAG, NumPy, Pandas, Scikit-Learn, Matplotlib, Agentic AI, Fine-Tuning
Databases: MySQL, PostgreSQL, MongoDB, PySpark, Kafka, Pinecone, Distributed Systems
Cloud Platforms: AWS (EC2, S3, Lambda, DynamoDB, SQS, ELB, KeyCloak), Docker, Kubernetes, CI/CD
Tools: Git, GitHub, VS Code

Work Experience

Software Engineer Intern — Feb 2025 – Aug 2025
Infosys — *Pune, India*

- Transformed a monolithic banking platform into reusable cloud-native REST microservices by migrating 15+ SOAP services with Java, Spring Boot, MongoDB, and AWS, applying object-oriented design principles and core data structures, to improve API response times by 35% while reducing maintenance effort by 50%.
- Automated CI/CD and containerized deployment pipelines using Jenkins, GitLab, Docker, and Kubernetes, achieving 85% deployment automation and 60% faster releases.

Projects

Natural Language-to-SQL Query Engine (Text2SQL) | https://github.com/asmipulgam/fieldcode

- ***Python, PostgreSQL, SQL, LangChain, Git, MCP, HTML5, CSS3***
- Engineered a production-style **GenAI** platform enabling 100+ users to query enterprise databases using natural language through **LLM-powered semantic reasoning** and secure backend APIs.
- Built scalable backend services using **Python**, integrating **Anthropic Claude API, LangChain**, and secure **MCP-based connectors** to translate natural language into optimized SQL queries across multiple data sources.
- Implemented **LangChain** query caching, optimized backend performance, and developed **PyTest**-based unit tests to validate query generation and API functionality, reducing average query response time by 40% while improving application reliability.

Agentic AI Financial Advisor Platform with RAG | sparkyfi.vercel.app

- ***Next.js, TypeScript, Pinecone, CSS, Vercel, RAG, Agentic AI***
- Developed a full-stack AI financial advisor, enabling users to analyze bank statements, insurance policies, leases, and loan documents through conversational AI and voice interaction.
- Architected a production-grade **RAG** and **agentic AI** pipeline with **Pinecone** vector search and semantic retrieval across 2,800+ State Farm and CFPB records, delivering personalized, context-aware financial recommendations.

Distributed Database Performance Optimization System | github.com/asmipulgam/Geo-Distributed-HealthCare-Database-System

- ***CockroachDB, MongoDB, Docker, SQL, Python, Distributed Systems***
- Designed a fault-tolerant distributed database platform supporting multi-region replication, high availability, and low-latency query execution using CockroachDB and MongoDB.
- Optimized query performance by refining replication, data placement, and consistency models, reducing cross-region latency by 30% and improving throughput by 25%.

Privacy-Preserving Language Model Fine-Tuning and Deployment | https://github.com/asmipulgam/Slm-Finetunning

- ***Python, PyTorch, Hugging Face Transformers, TinyLlama, Qwen2.5, LoRA, QLoRA***
- Built an on-device hate speech detection system using **TinyLlama, Qwen2.5, PyTorch, and Hugging Face Transformers**, ensuring 100% on-device inference without transmitting sensitive user data.
- Enhanced Small Language Model training efficiency using **LoRA** and **QLoRA**, achieving 2-hour **fine-tuning** on a 4GB GPU and increasing offensive-content recall from 39% to 55%.

Leadership Experience

Technical Lead — Aug 2023 – May 2024
Google Developer Student Club (GDSC)

- Led technical workshops and speaker sessions, guiding students through web development, cloud services, data structures and algorithms, and AI/ML fundamentals.

Hackathon Team Lead — April 2026
ASU Innovation Hacks

- Led a team in developing Sparkify, an agentic AI financial advisor, coordinating technical decisions, task execution, and rapid prototyping throughout the hackathon.